Tutorial 3: Bayesian Estimation and Regularization

Three required tasks and one optional challenge

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How to use this tutorial

- Attempt each problem before reading its solution.
- Write the main calculation or argument clearly.
- Explain what the result means, not only the final number or formula.
- Complete Tasks 1–3 first. The final task is an optional challenge.

Task 1: Beta updating

We observe four Bernoulli trials:

$$m = 3, \qquad n = 4.$$

Consider two priors:

$$\text{Prior A: } \mathrm{Beta}(2, 2), \qquad \text{Prior B: } \mathrm{Beta}(20, 20).$$

Both priors have mean 0.5, but Prior B is much more concentrated.

1. Find the posterior distribution under each prior.
2. Compute the posterior mean and next-trial success probability for each prior. Compare them with the MLE.
3. For Prior A, compute the MAP estimate. Explain why it differs from the posterior mean.

Solution 1: Posterior update

1. Find the two posterior distributions.
Beta–Bernoulli updating gives

$$\mathrm{Beta}(\alpha, \beta) \to \mathrm{Beta}(\alpha + m, \beta + n - m).$$

Since $m = 3$ and $n - m = 1$, Prior A gives

$$\mathrm{Beta}(2, 2) \to \boxed{\mathrm{Beta}(5, 3)}.$$

Prior B gives

$$\mathrm{Beta}(20, 20) \to \boxed{\mathrm{Beta}(23, 21)}.$$

Both priors start at mean 0.5, but the stronger prior changes less after the same four observations.

Solution 1: Estimates

2. Compute posterior means, predictions, and MLE.
For Prior A,

$$\mathbb{E}[\pi \mid \mathcal{D}] = P(x_* = 1 \mid \mathcal{D}) = \frac{5}{8} = 0.625.$$

For Prior B,

$$\mathbb{E}[\pi \mid \mathcal{D}] = P(x_* = 1 \mid \mathcal{D}) = \frac{23}{44} \approx 0.523.$$

The MLE uses only the observed data:

$$\hat{\pi}_{\mathrm{ML}} = \frac{3}{4} = 0.75.$$

Prior B keeps the estimate much closer to 0.5 because it represents stronger prior information.

Solution 1: MAP

3. Compute MAP for Prior A and compare it with the posterior mean.
Under Prior A,

$$\pi \mid \mathcal{D} \sim \mathrm{Beta}(5, 3).$$

Both shape parameters exceed one, so the MAP estimate is

$$\hat{\pi}_{\mathrm{MAP}} = \frac{5-1}{5+3-2} = \frac{2}{3} \approx 0.667.$$

The posterior mean is

$$\mathbb{E}[\pi \mid \mathcal{D}] = 0.625.$$

They differ because MAP selects the highest-density point, whereas the posterior mean averages over the entire posterior distribution.

The same posterior can produce different point summaries depending on the question being asked.

Task 2: Gaussian updating

Suppose our prior belief about an unknown mean is

$$\mu \sim \mathcal{N}(10, 4).$$

We observe $n = 2$ measurements with

$$\bar{x} = 14, \qquad \sigma^2 = 8,$$

where σ^2 is the known observation variance.

1. Compute the prior precision and data precision. Which source receives more weight?
2. Compute the posterior mean and posterior variance.
3. Let x_* be a new observation from the same population. Find its predictive mean and variance, and explain why the predictive variance is larger.

Solution 2: Precision

1. Compute and compare the two precisions.

The prior precision is

$$\lambda_0 = \frac{1}{4} = 0.25.$$

The data precision is

$$\lambda_D = \frac{n}{\sigma^2} = \frac{2}{8} = 0.25.$$

The precisions are equal, so the prior mean 10 and sample mean 14 receive equal weight.

Solution 2: Posterior

2. Compute the posterior mean and variance.

$$\mu_n = \frac{0.25(10) + 0.25(14)}{0.25 + 0.25} = 12.$$

The posterior precision is

$$\lambda_n = 0.25 + 0.25 = 0.5,$$

so

$$\tau_n^2 = \frac{1}{0.5} = 2.$$

Therefore,

$$\boxed{\mu \,|\, \mathcal{D} \sim \mathcal{N}(12, 2)}.$$

Solution 2: Prediction

3. Find the predictive distribution for a new observation. A new observation satisfies

$$x_* \mid \mu \sim \mathcal{N}(\mu, 8).$$

The predictive mean is

$$\mathbb{E}[x_* \mid \mathcal{D}] = \mu_n = 12.$$

The predictive variance contains two sources:

$$\mathrm{Var}(x_* \mid \mathcal{D}) = \underbrace{8}_{\text{observation noise}} + \underbrace{2}_{\text{uncertainty about } \mu} = 10.$$

Hence,

$$\boxed{x_* \mid \mathcal{D} \sim \mathcal{N}(12, 10)}.$$

The predictive variance is larger than the posterior variance because predicting a new observation includes both uncertainty about μ and new observation noise.

Task 3: MAP and ridge

In the lecture, ridge regression used a zero-centered Gaussian prior. Now suppose instead

$$\boldsymbol{w} \sim \mathcal{N}(\boldsymbol{w}_0, \tau^2 \boldsymbol{I}),$$

while the regression model is

$$\boldsymbol{y} \mid \boldsymbol{w} \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{w}, \sigma^2 \boldsymbol{I}).$$

1. Derive the MAP objective up to constants. What replaces the usual $\|\boldsymbol{w}\|_2^2$ penalty?
2. Differentiate the objective and derive the normal equations.
3. Explain what happens when $\boldsymbol{w}_0 = \boldsymbol{0}$ and when $\tau^2 \to \infty$.

Solution 3: MAP objective

1. Derive the MAP objective.
The Gaussian likelihood contributes

$$-\log p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + C.$$

The Gaussian prior contributes

$$-\log p(\boldsymbol{w}) = \frac{1}{2\tau^2}\|\boldsymbol{w} - \boldsymbol{w}_0\|_2^2 + C.$$

Therefore MAP minimizes

$$\frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \frac{1}{2\tau^2}\|\boldsymbol{w} - \boldsymbol{w}_0\|_2^2.$$

After multiplying by $2\sigma^2$,

$$\boxed{\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \lambda\|\boldsymbol{w} - \boldsymbol{w}_0\|_2^2}, \qquad \lambda = \frac{\sigma^2}{\tau^2}.$$

The penalty now shrinks weights toward $\boldsymbol{w}_0$, not toward zero.

Solution 3: Normal equations

2. Differentiate and derive the normal equations.
Differentiating the objective gives

$$-2\boldsymbol{X}^\top(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{w})+2\lambda(\boldsymbol{w}-\boldsymbol{w}_0)=\boldsymbol{0}.$$

Rearranging,

$$\boxed{(\boldsymbol{X}^\top\boldsymbol{X}+\lambda\boldsymbol{I})\hat{\boldsymbol{w}}_{\mathrm{MAP}}=\boldsymbol{X}^\top\boldsymbol{y}+\lambda\boldsymbol{w}_0}.$$

Compared with ordinary ridge, the prior center $\boldsymbol{w}_0$ contributes the additional term

$$\lambda\boldsymbol{w}_0.$$

Solution 3: Special cases

3(a). What happens when $w_0 = 0$?
The penalty becomes

$$\|\boldsymbol{w} - \boldsymbol{w}_0\|_2^2 = \|\boldsymbol{w}\|_2^2,$$

so we recover standard ridge regression.

3(b). What happens when $\tau^2 \to \infty$?
Since

$$\lambda = \frac{\sigma^2}{\tau^2},$$

we obtain

$$\tau^2 \to \infty \quad \Rightarrow \quad \lambda \to 0.$$

The prior becomes weak, the penalty disappears, and MAP approaches ordinary least squares.

Optional: Ridge experiment

Write a Python program to study how ridge regression behaves when two features are highly correlated. Generate a synthetic dataset with

$$n = 80, \qquad \boldsymbol{w}_{\text{true}} = (2, 2)^\top.$$

Generate x_1 from a standard Gaussian distribution and construct

$$x_2 = x_1 + \text{small Gaussian noise},$$

so that the two features are almost identical. Generate the target using

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{w}_{\text{true}} + \text{Gaussian noise}.$$

1. Implement ordinary least squares and ridge regression with $\lambda = 1$.
2. Compare their fitted coefficients and training MSE.
3. Repeat the experiment 100 times with new response noise and compare the variability of the fitted coefficients.
4. Explain why ridge produces more stable coefficients.

Solution: Data generation

1. Generate a dataset with highly correlated features. One possible implementation is:

```
import numpy as np

rng = np.random.default_rng(7)
n = 80

x1 = rng.normal(size=n)
x2 = x1 + 0.02 * rng.normal(size=n)
X = np.column_stack([x1, x2])

w_true = np.array([2.0, 2.0])
y = X @ w_true + 0.5 * rng.normal(size=n)
```

Because x_2 differs from x_1 only by small noise,

$$x_2 \approx x_1.$$

The two columns of $\boldsymbol{X}$ therefore contain almost the same information.

Solution: OLS and ridge

1. Fit ordinary least squares and ridge regression.

```
lam = 1.0

w_ols = np.linalg.lstsq(X, y, rcond=None)[0]

w_ridge = np.linalg.solve(
    X.T @ X + lam * np.eye(2),
    X.T @ y
)
```

With the fixed random seed, the fitted coefficients are approximately

$$\hat{\boldsymbol{w}}_{\mathrm{OLS}} = (4.16, -0.22)^{\top},$$

while

$$\hat{\boldsymbol{w}}_{\mathrm{ridge}} = (1.98, 1.93)^{\top}.$$

Both models can make similar predictions even though their individual coefficients are very different.

Solution: Training error

2. Compare the training MSE.
Compute

```
mse_ols = np.mean((y - X @ w_ols)**2)
mse_ridge = np.mean((y - X @ w_ridge)**2)

print(mse_ols, mse_ridge)
```

For this dataset,

$$\mathrm{MSE}_{\mathrm{OLS}} \approx 0.215, \qquad \mathrm{MSE}_{\mathrm{ridge}} \approx 0.217.$$

The prediction errors are almost the same, even though

$$(4.16, -0.22) \qquad \text{and} \qquad (1.98, 1.93)$$

are very different coefficient estimates.

Good training fit does not guarantee stable or interpretable coefficients.

Solution: Repeated trials

3. Repeat the fit with new response noise.
Keep $\boldsymbol{X}$ fixed and regenerate only $\boldsymbol{y}$:

```
rng2 = np.random.default_rng(17)
ols_coef = []
ridge_coef = []

for _ in range(100):
    y_new = X @ w_true + 0.5 * rng2.normal(size=n)

    ols_coef.append(
        np.linalg.lstsq(X, y_new, rcond=None)[0]
    )
    ridge_coef.append(
        np.linalg.solve(
            X.T @ X + np.eye(2),
            X.T @ y_new
        )
    )
```

Solution: Variability

3. Compare coefficient variability.
Convert the results to arrays and compute their standard deviations:

```
ols_coef = np.array(ols_coef)
ridge_coef = np.array(ridge_coef)

print(ols_coef.std(axis=0))
print(ridge_coef.std(axis=0))
```

The standard deviations are approximately

$$\mathrm{SD}(\hat{\boldsymbol{w}}_{\mathrm{OLS}}) = (2.96, 2.96),$$

compared with

$$\mathrm{SD}(\hat{\boldsymbol{w}}_{\mathrm{ridge}}) = (0.050, 0.047).$$

OLS coefficients vary dramatically across repeated noisy datasets, while ridge coefficients remain much more stable.

Solution: Interpretation

4. Why is ridge more stable?
Because

$$x_1 \approx x_2,$$

the data cannot clearly determine how the total effect should be divided between the two coefficients. OLS can therefore increase one coefficient and decrease the other while keeping nearly the same predictions. Ridge adds the penalty

$$\lambda \|\boldsymbol{w}\|_2^2,$$

which discourages unnecessarily large coefficients. This selects a more balanced and less noise-sensitive solution.

Ridge can greatly reduce coefficient variability even when its training error is almost unchanged.

Tutorial 3 complete

Questions and discussion